BIS 629: Stepped Wedge Designs

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Motivating example

Postpartum intrauterine device (PPIUD) study design

- Collaboration with
  - International Federation of Gynaecology and Obstetrics (FIGO)
  - Association of Gynaecologists and Obstetricians of Tanzania (AGOTA)
  - Harvard T.H. Chan School of Public Health (HSPH)

- Impact of PPIUD use in Tanzania



Motivating example

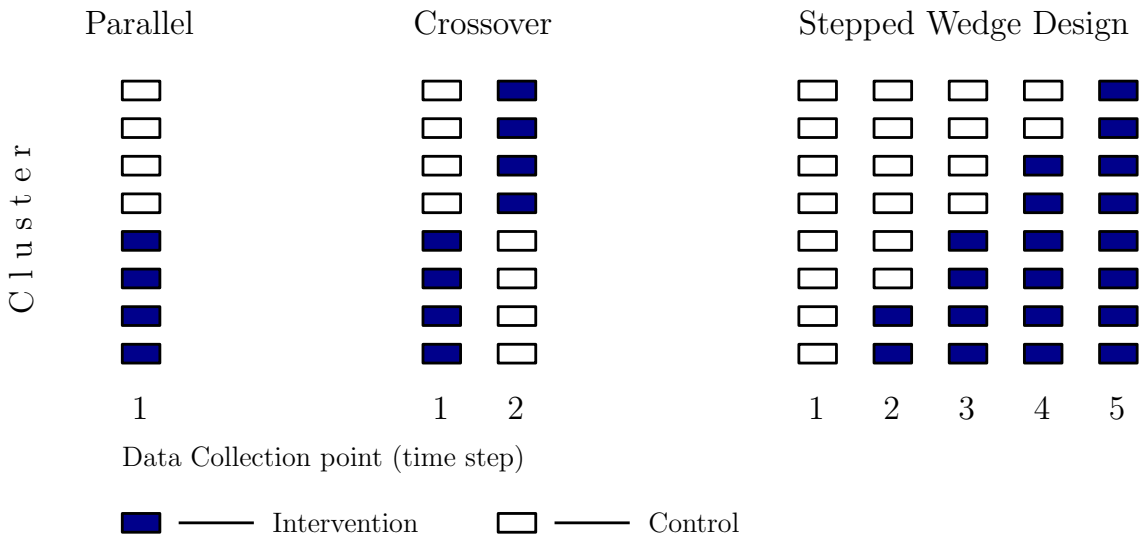
(Canning et al, 2016[1])

- During the year following the birth of a child, 40% of women are estimated to have an unmet need for contraception.
- Postpartum intrauterine device (PPIUD) meets women's need for contraceptive protection following childbirth
  - Long-term contraceptive protection
  - Reversible
  - Safe, effective and convenient
  - Does not interfere with breastfeeding during the postpartum period
- Uptake of PPIUD is low
  - PPIUD insertion should be performed by a trained provider in the early postpartum period
  - The services are not widely available
- The International Federation of Gynaecology and Obstetrics (FIGO) implemented an intervention that institutionalize PPIUD services
  - as a routine part of antenatal counselling
  - as a part of delivery room services

[1] Canning, D., Shah, I. H., Pearson, E., Pradhan, E., Karra, M., Senderowicz, L.,Bärnighausen T, Spiegelman D, & Langer, A. (2016). Institutionalizing postpartum intrauterine device (IUD) services in Sri Lanka, Tanzania, and Nepal: study protocol for a cluster-randomized stepped-wedge trial. BMC pregnancy and childbirth, 16(1), 1-11.

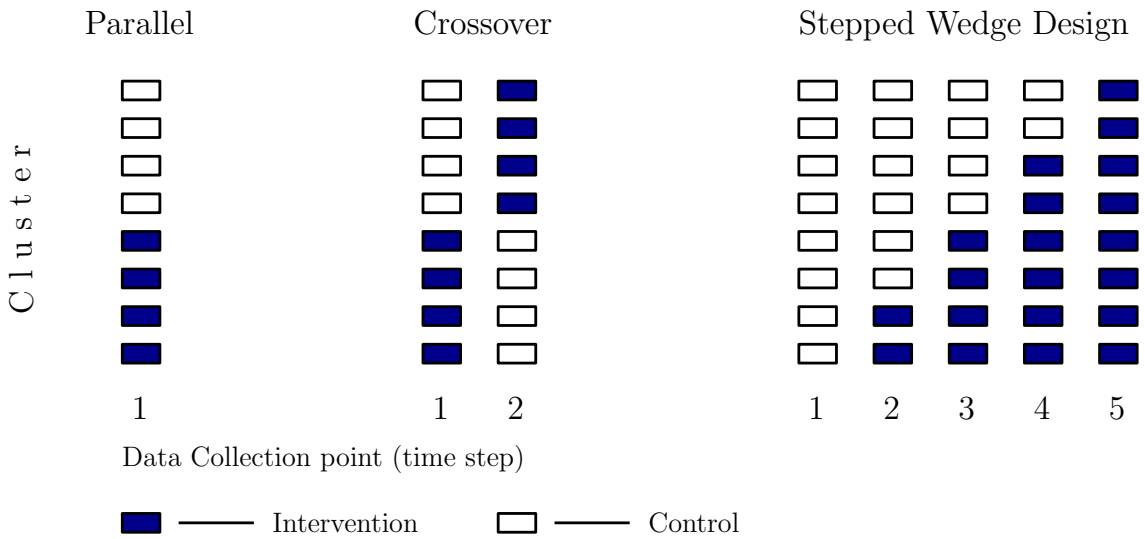
Cluster randomized trials

- Different types of cluster randomized trials (CRT)

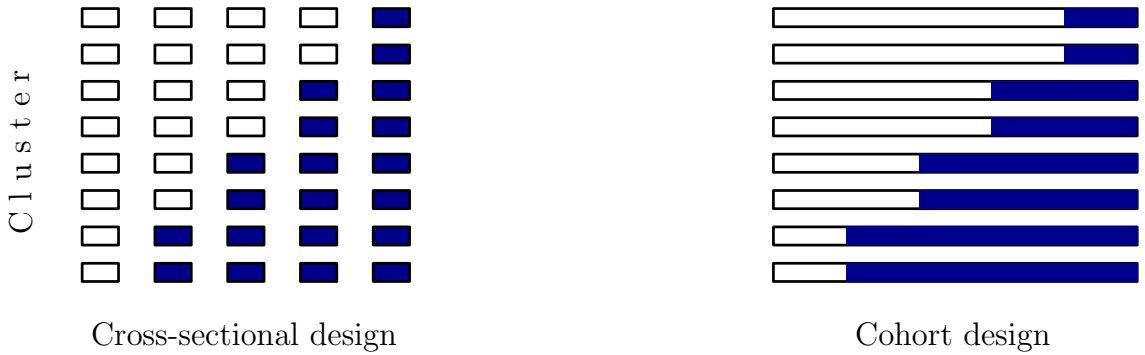


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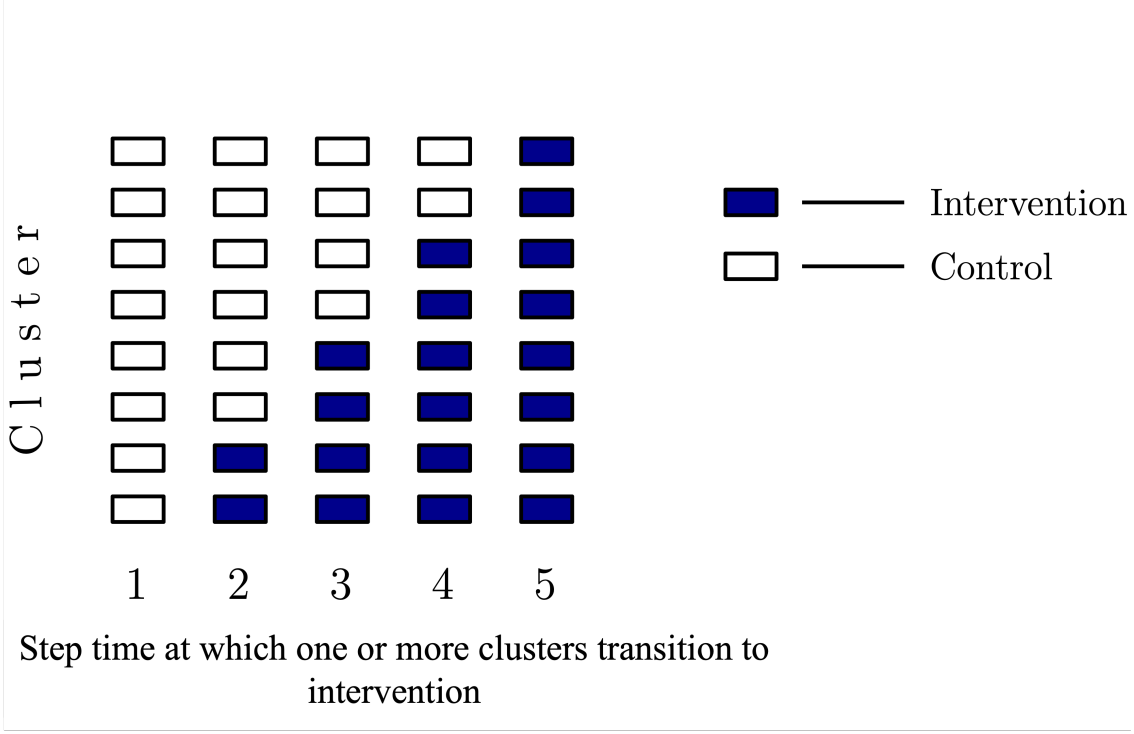


- Cross-sectional v.s. cohort SWDs



Stepped wedge design

- Stepped Wedge Design (SWD)

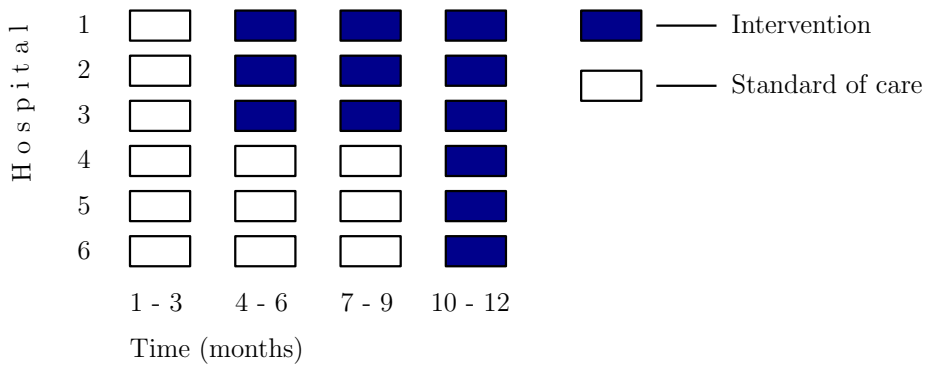


- Ethics: all clusters need to get the potentially effective intervention
- Feasibility: not able to give the intervention simultaneously to many clusters

Motivating example

Postpartum intrauterine device (PPIUD) study design

- Power to detect the effectiveness of PPIUD in preventing unintended pregnancy for 1.5 years following the index birth
- Enrollment period is one year long.
- There are $K = 6$ clusters (hospitals), and $T = 4$ time steps.
- Each step is 3 months long for enrollment.
- m_T = Approximately 900 women join each hospital each three month step
- The cluster size is expected to be $m = Tm_T$ =900*4=3,600.



Original power calculation method

Hussey and Hughes (2007)[1]

- Apply a linear mixed model (LMM) for continuous outcomes

$$Y_{ktj} = \beta_0 + X_{ktj}\beta^* + \gamma_t + b_k + \epsilon_{ktj},$$

where Y_{ktj} is the outcome of individual $j(\in \{1, \cdots, m_T\})$ at step $t(\in \{1, \cdots, T\})$ from cluster $k(\in \{1, \cdots, K\})$,

X_{ktj} is an indicator of the treatment (1=intervention; 0=control),

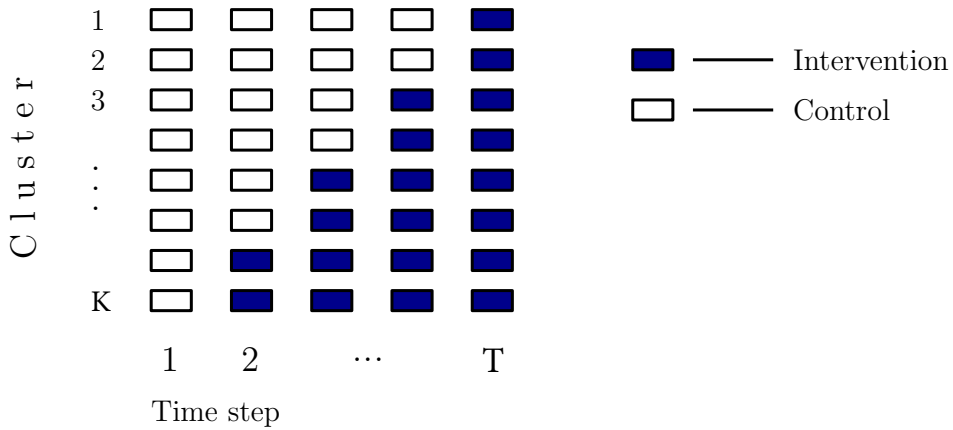
β_0 is the expectation of the outcome at baseline (control at step 1),

β^* is the treatment effect,

γ_t is the time effect corresponding to step t ($\gamma_1 = 0$ for identifiability),

$\epsilon_{ktj} \sim N(0, \sigma^2_W)$ is the error term,

$b_k \sim N(0, \sigma^2_B)$ is the random effect for cluster k



[1] Hussey MA, Hughes JP. Design and analysis of stepped wedge cluster randomized trials. Contemp Clin Trials. 2007; 28(2):182-91.

Original SWD power calculation method

Hussey and Hughes (2007)

- Power calculation is based on a Wald test for

$$H_0 : \beta^* = 0 \text{ versus } H_A : \beta^* = \beta_A.$$

Original SWD power calculation method

Hussey and Hughes (2007)

- Power calculation is based on a Wald test for

$$H_0 : \beta^* = 0 \text{ versus } H_A : \beta^* = \beta_A.$$

- The asymptotic power for conducting a two-tailed test of size α is

$$power = \Phi\left(\frac{|\beta_A|}{\sqrt{Var(\hat{\beta}^*)}} - Z_{1-\alpha/2}\right),$$

where $\Phi(\cdot)$ is the cumulative standard Normal distribution,
$Z_{1-\alpha/2}$ is the $(1-\alpha/2)$th quantile of the standard Normal distribution.

Existing method

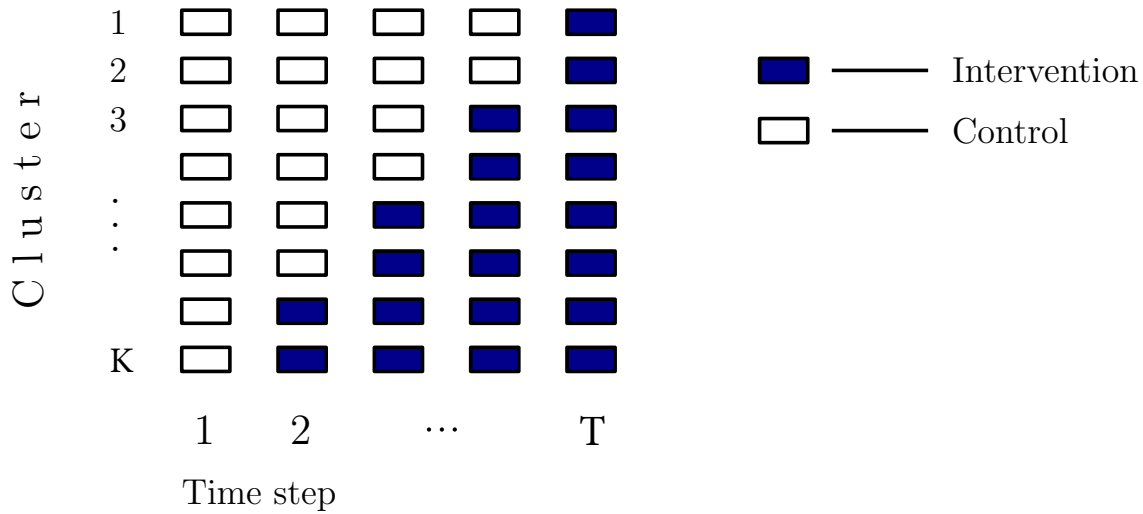
Hussey and Hughes (2007)

- Pros
  - closed-form formula for power calculation
  - Simple and fast
- Cons
  - Assumes σ^2_W and σ^2_B are known
  - An approximation for binary outcomes

Method – a Maximum Likelihood Approach to estimation and inference under GLMM

(Zhou et al, 2020[1])

- We consider a binary intervention X_{ktj} and a binary outcome Y_{ktj} for individual j in cluster k at step t, where $k \in \{1, \cdots, K\}$, $t \in \{1, \cdots, T\}$, and $j \in \{1, \cdots, m_T\}$.
- The cluster size is $m = Tm_T$, and the total sample size is $n = KTm_T$.
- Two cases: without time effects and with time effects.



[1]Xin Zhou, Xiaomei Liao, Lauren M Kunz, Sharon-Lise T Normand, Molin Wang, Donna Spiegelman. (2020). A maximum likelihood approach to power calculations for stepped wedge designs of binary outcomes. Biostatistics, 21(1), 102-121.

Method – a Maximum Likelihood Approach for cross-sectional SWDs, subject-specific model

The model without time effects

- Apply a generalized linear mixed model (GLMM) for binary outcomes

$$g(p_{ktj}) = \beta_0 + X_{ktj}\beta^* + b_k, \tag{1}$$

where β_0 is the probability of the outcome at baseline (control at step 1),
β^* is the intervention effect
$b_k \sim N(0, \sigma_B^2)$ is the random effect for cluster k,
$p_{ktj} = E(Y_{ktj}|X_{ktj}, b_k) = Pr(Y_{ktj} = 1|X_{ktj}, b_k)$

- Link function $g(\cdot)$
  - Identity link – parameter of interest is the risk difference
  - Log link – parameter of interest is the risk ratio (relative risk)
  - **Logit link – parameter of interest is odds ratio**

Method – a Maximum Likelihood Approach

Power calculation

- The asymptotic power for $H_0 : \beta^* = 0$ versus $H_A : \beta^* = \beta^*_A$

$$power = \Phi\left(\frac{|\beta^*_A|}{\sqrt{Var(\hat{\beta}^*)}} - Z_{1-\alpha/2}\right).$$

From likelihood theory, $Var(\hat{\beta}^*)$ can be obtained by the inverse of information matrix.

Method – a Maximum Likelihood Approach

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From likelihood theory, $Var(\hat{\beta}^*)$ can be obtained by the inverse of information matrix.

- Reminder: Likelihood theory
  - Let $\boldsymbol{\theta}$ be an arbitrary p-dimensional vector of parameters, one element of which is the parameter of interest
  - Score function: $\boldsymbol{U}(\boldsymbol{\theta}) = \frac{\partial \ell(\boldsymbol{\theta};\boldsymbol{y})}{\partial \boldsymbol{\theta}}$.
  - Information matrix: $\boldsymbol{I}(\boldsymbol{\theta}) = E(\boldsymbol{U}(\boldsymbol{\theta})\boldsymbol{U}(\boldsymbol{\theta})^T) = \frac{\partial \ell^2(\boldsymbol{\theta};\boldsymbol{y})}{\partial \boldsymbol{\theta}^2} = \frac{\partial \boldsymbol{U}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$.
  - Asymptotic distribution of MLE

$$\hat{\boldsymbol{\theta}}_{ML} \Rightarrow N_p(\boldsymbol{\theta}, \boldsymbol{I}(\boldsymbol{\theta})^{-1})$$

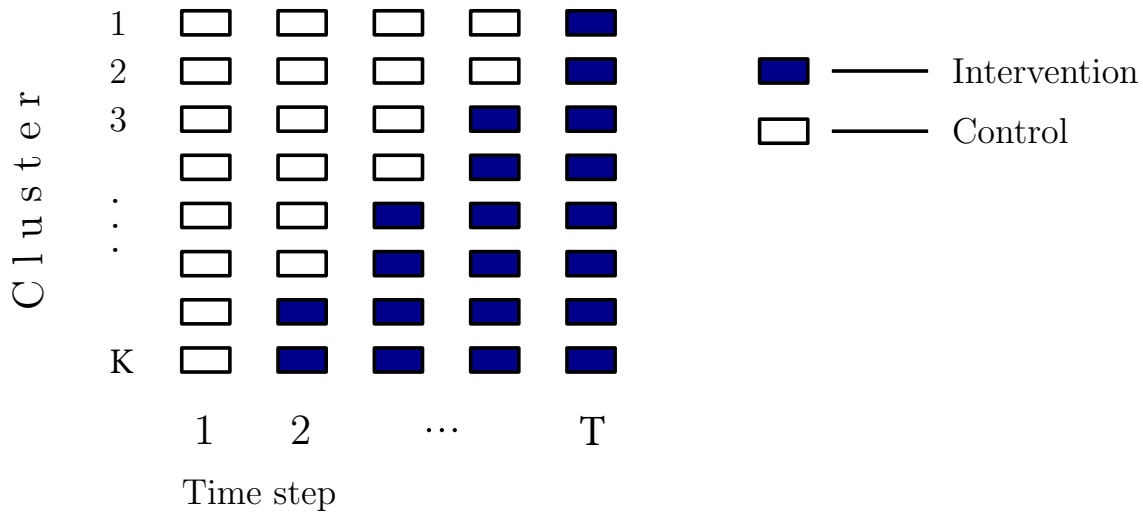
Method – a Maximum Likelihood Approach

For the model without time effects

- Data for a single cluster can be re-organized as $\boldsymbol{Z}_k = (Z_{k00}, Z_{k01}, Z_{k10}, Z_{k11})$:

Intervention, X_{ktj}	Outcome, Y_{ktj} = 0	Outcome, Y_{ktj} = 1	
0	Z_{k00}	Z_{k01}	$Z_{k0.}$
1	Z_{k10}	Z_{k11}	$Z_{k1.}$
			m

where $Z_{k0.}$ and $Z_{k1.}$ are fixed by the SWD. In a balanced SWD, $Z_{k0.} = Z_{k1.} = KTm_T$.



Method – a Maximum Likelihood Approach

- For the logit link, Model (1) is rewritten as

$$p_{ktj} = g^{-1}(\beta_0 + X_{ktj}\beta^* + b_k) = g^{-1}(\eta), \qquad t = 1, \cdots, T, \quad j = 1, \cdots, m_T. \tag{2}$$

  where $g^{-1}(\cdot)$ is the inverse link function.
- The inverse logit link function is

$$g^{-1}(\eta) = \frac{exp(\eta)}{1 + exp(\eta)}.$$

Method – a Maximum Likelihood Approach

- For the logit link, Model (1) is rewritten as

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$$g^{-1}(\eta) = \frac{exp(\eta)}{1 + exp(\eta)}.$$

- The observed data likelihood for parameters $\boldsymbol{\theta} = (\beta_0, \beta^*, \sigma_B^2)$ is

$$L(\boldsymbol{\theta}) = \prod_{k=1}^{K} \int \binom{Z_{k0\cdot}}{Z_{k00}} \left[1 - g^{-1}(\beta_0 + b_k)\right]^{Z_{k00}} \left[g^{-1}(\beta_0 + b_k)\right]^{Z_{k01}} \cdot$$
$$\binom{Z_{k1\cdot}}{Z_{k10}} \left[1 - g^{-1}(\beta_0 + \beta^* + b_k)\right]^{Z_{k10}} \left[g^{-1}(\beta_0 + \beta^* + b_k)\right]^{Z_{k11}} f(b_k) db_k,$$

  where $f(b_k)$ is the density function of b_k, assumed normal for now
- If need be, the density function of b_k is suitably modified from a N(0,1) to ensure that all values of $g^{-1}(\cdot)$ are probabilities, i.e. range between [0,1], and the density integrates to 1.

Method – a Maximum Likelihood Approach

- When the random effects are assumed to have mean 0, constant variance and no other parameters describing their distribution, the parameters of the likelihood are $\boldsymbol{\theta} = (\beta_0, \beta^*, \sigma^2_B)$.
- The log-likelihood $\ell(\boldsymbol{\theta})$ is

$$\ell(\boldsymbol{\theta}) = \sum_{k=1}^{K} \int \left[1 - g^{-1}(\beta_0 + b_k)\right]^{Z_{k00}} \left[g^{-1}(\beta_0 + b_k)\right]^{Z_{k01}} \cdot$$

$$\binom{Z_{k1\cdot}}{Z_{k10}} \left[1 - g^{-1}(\beta_0 + \beta^* + b_k)\right]^{Z_{k10}} \left[g^{-1}(\beta_0 + \beta^* + b_k)\right]^{Z_{k11}} f(b_k) db_k,$$

- When g is the logit link, we often assume that $f(b_k)$ is N(0,1) and the limits of integration are $-\infty, \infty$.
- When g is the identity link and $f(b_k)$ is assumed to be normal, there are restrictions on the acceptable values of b_k to ensure that $1 \geq \beta_0 + \beta^* + b_k \geq 0$. Thus, the lower limit of integration, $v(\boldsymbol{\theta}) = -\beta_0$ and the upper limit of integration, $u(\boldsymbol{\theta}) = 1 - \beta_0$ for control periods, and $v(\boldsymbol{\theta}) = -\beta_0 - \beta^*$ and $u(\boldsymbol{\theta}) = 1 - \beta_0 - \beta^*$ for intervention periods. In addition, $f(b_k)$ needs to be adjusted as a truncated normal to ensure that it is a well defined probability function that integrates to one, i.e.

$$f(b_k) = e^{\frac{-b_k^2}{2\sigma^2_B}} \left[\frac{I(0 \leq \beta^* \leq 1)}{\int_{-\beta_0}^{1-\beta_0-\beta^*} e^{-\frac{b_2}{2\sigma^2_B}} db} + \frac{I(-1 \leq \beta^* \leq 0)}{\int_{-\beta_0-\beta^*}^{1-\beta_0} e^{-\frac{b_2}{2\sigma^2_B}} db} \right]$$

, where $I(\cdot)$ is the indicator function

- We use the Leibniz rule to obtain the derivative of the log-likelihood function, which is an integral function
- The Leibniz rule for differentiation of a definite integral is

$$\frac{\partial}{\partial \boldsymbol{\theta}} \left(\int_{v(\boldsymbol{\theta})}^{u(\boldsymbol{\theta})} f(\boldsymbol{\theta}, b_k) db_k \right) = f(\boldsymbol{\theta}, u(\boldsymbol{\theta})) \frac{\partial u(\boldsymbol{\theta})}{\partial \boldsymbol{\theta})} - f(\boldsymbol{\theta}, v(\boldsymbol{\theta})) \cdot \frac{\partial v(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} + \int_{v(\boldsymbol{\theta})}^{u(\boldsymbol{\theta})} \frac{\partial f(\boldsymbol{\theta}, b_k)}{\partial \boldsymbol{\theta}} db_k$$

Method – a Maximum Likelihood Approach

Asymptotically, the theoretical variance of $\hat{\theta}_{ML}$ is given by

$$
\begin{aligned}
Var(\hat{\theta}_{ML}) &= [I(\boldsymbol{\theta})]^{-1} = \left(\sum_{k=1}^{K} E_{\boldsymbol{Z}_k} \left[\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right)_{3\times 1} \left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right)^{T}_{1\times 3} \right] \right)^{-1} \\
&= \left(\sum_{k=1}^{K} \left[\sum_{Z_{k10}=0}^{Z_{k1\cdot}} \sum_{Z_{k00}=0}^{Z_{k0\cdot}} \left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right) \left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right)^{T} Pr(\boldsymbol{Z}_k) \right] \right)^{-1},
\end{aligned}
$$

where, as above, $Pr(\mathbf{Z}_k)$ is given by

$$
\begin{aligned}
Pr(\mathbf{Z}_k) = \int & \binom{Z_{k0\cdot}}{Z_{k00}} \left[1 - g^{-1}(\beta_0 + b_i) \right]^{Z_{k00}} \left[g^{-1}(\beta_0 + b_k) \right]^{Z_{k01}} \cdot \\
& \binom{Z_{k1\cdot}}{Z_{k10}} \left[1 - g^{-1}(\beta_0 + \beta^* + b_k) \right]^{Z_{k10}} \left[g^{-1}(\beta_0 + \beta^* + b_k) \right]^{Z_{k11}} f(b_k) db_k .
\end{aligned}
$$

Method – a Maximum Likelihood Approach

In summary,

- Calculate information matrix

$$\begin{aligned} \boldsymbol{I}(\boldsymbol{\theta}) &= \sum_{k=1}^{K} E_{\boldsymbol{Z}_k}\left[\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)^T\right] \\ &= \sum_{k=1}^{K}\left[\sum_{Z_{k10}=0}^{Z_{k1\cdot}}\sum_{Z_{k00}=0}^{Z_{k0\cdot}}\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)\left(\frac{\partial \ell_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)^T Pr(\boldsymbol{Z}_k)\right] \end{aligned}$$

- Calculate $Var(\hat{\boldsymbol{\theta}}) = \boldsymbol{I}(\boldsymbol{\theta})^{-1}$, and $Var(\hat{\beta}^*)$ is the (2,2)-element of $Var(\hat{\boldsymbol{\theta}})$
- Power calculation

$$power = \Phi\left(\frac{|\beta_A|}{\sqrt{Var(\hat{\beta}^*)}} - Z_{1-\alpha/2}\right)$$

Method – a Maximum Likelihood Approach

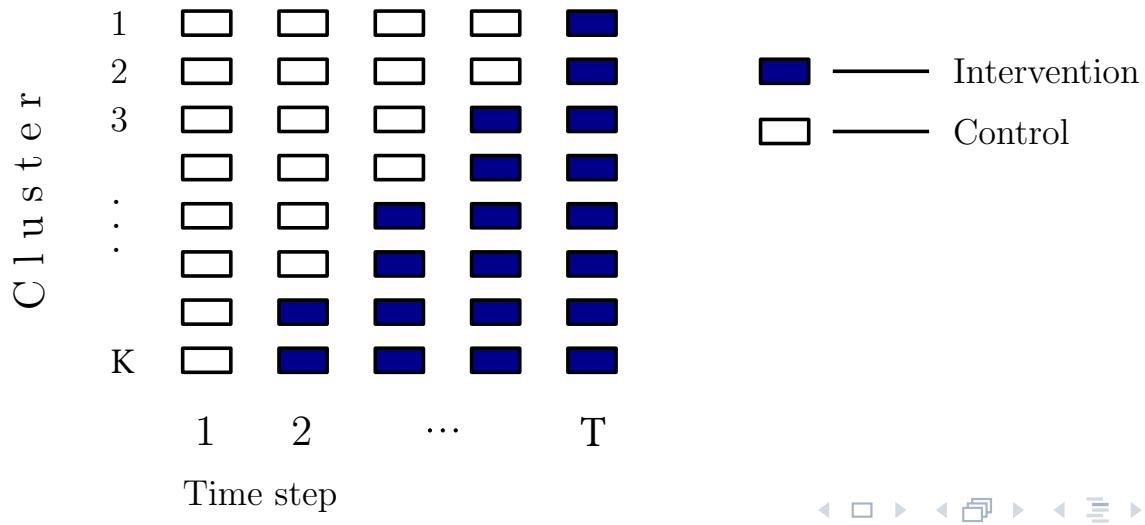
The model with time effects

- Apply a generalized linear mixed model (GLMM) using the logistic link for binary outcomes

$$p_{ktj} = g^{-1}(\beta_0 + X_{ktj}\beta^* + \gamma_t + b_k),$$

where γ_t is the time effect corresponding to step t ($\gamma_1 = 0$ for identifiability) and $g(\beta_0 + X_{ktj}\beta^* + \gamma_t + b_k) = \frac{e^{\beta_0 + X_{ktj}\beta^* + \gamma_t + b_k)}}{1 + e^{\beta_0 + X_{ktj}\beta^* + \gamma_t + b_k}}$.

- The parameters $\boldsymbol{\theta} = (\beta_0, \beta^*, \gamma_2, \cdots, \gamma_T, \sigma_B^2)$.
- Data for cluster k at step t are summarized as $\boldsymbol{Z}_{kt} = (Z_{kt0}, Z_{kt1})$, where Z_{kt0} — # of individuals with outcome 0,
  Z_{kt1} — # of individuals with outcome 1.



Method – a Maximum Likelihood Approach

- The observed data log-likelihood function is

$$\begin{aligned}\ell(\boldsymbol{\theta}) &= \sum_{k=1}^{K} \log \left[\int \left\{ \prod_{t=1}^{T} \binom{m_T}{Z_{kt0}} \left(g^{-1}(\beta_0 + X_{kt}\beta^* + \gamma_t + b_k) \right)^{Z_{kt1}} \right. \right. \\ &\qquad \left. \left. \left(1 - g^{-1}(\beta_0 + X_{kt}\beta^* + \gamma_t + b_k) \right)^{Z_{kt0}} \right\} f(b_k) db_k \right] \\ &= \sum_{k=1}^{K} \ell_k(\boldsymbol{\theta})\end{aligned}$$

- When g is the logit link, the density function of b_k will again be assumed to be a N(0,1)

$$f(b_k) = \frac{1}{\sqrt{2\pi\sigma_B^2}} \exp\left(-\frac{b^2}{2\sigma_B^2} \right).$$

- When g is the identity link, as in Zhou et al. 2020 paper, a truncated normal is needed to ensure probabilities $\in [0, 1]$ and the density integrates to 1

$$f(b_k) = e^{\frac{-b_k^2}{2\sigma_B^2}} \frac{\Pi_{t=1}^{T} I(0<\beta_0+\beta^*+\gamma_t+b_k<1) I(0<\beta_0+\gamma_t+b_k<1)}{\int_{-\infty}^{\infty} \Pi_{t=1}^{T} I(0<\beta_0+\beta^*+\gamma_t+b<1) I(0<\beta_0+\gamma_t+b<1) e^{-\frac{b_2}{2\sigma_B^2}} db}$$

Method – a Maximum Likelihood Approach

- Recall: data for cluster k at step t are summarized as $\boldsymbol{Z}_{kt} = (Z_{kt0}, Z_{kt1})$,
- Data for cluster k: $\boldsymbol{Z}_k = (\boldsymbol{Z}_{k1}, \cdots, \boldsymbol{Z}_{kT})$.
- Information matrix

$$\begin{aligned} \boldsymbol{I}(\boldsymbol{\theta}) &= \sum_{k=1}^{K} E_{\boldsymbol{Z}_k}\left[\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)^T\right] \\ &= \sum_{k=1}^{K}\left[\sum_{Z_{k10}=0}^{m_T}\cdots\sum_{Z_{kT0}=0}^{m_T}\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)^T Pr(\boldsymbol{Z}_k)\right], \end{aligned}$$

where $Pr(\mathbf{Z}_k)$ is given by

$$\begin{aligned} Pr(\mathbf{Z}_k) = \int \Bigg\{ &\prod_{t=1}^{T}\binom{m_T}{Z_{kt0}}\left(g^{-1}(\beta_0 + X_{kt}\beta^* + \gamma_t + b_k)\right)^{Z_{kt1}} \\ &\left(1 - g^{-1}(\beta_0 + X_{kt}\beta^* + \gamma_t + b_k)\right)^{Z_{kt0}}\Bigg\} f(b_k)db_k. \end{aligned}$$

Method – a Maximum Likelihood Approach

Again, a similar procedure for the model with time effects

- Calculate information matrix

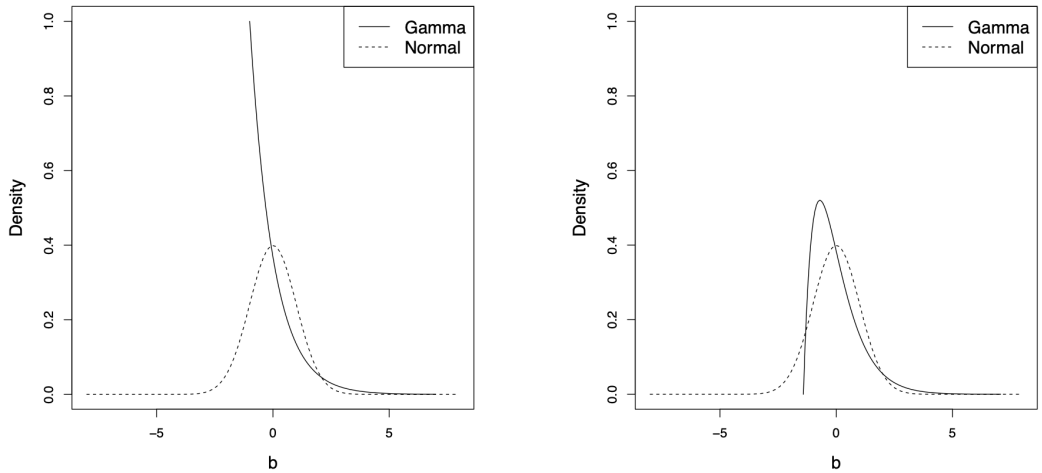
$$\begin{aligned} \boldsymbol{I}(\boldsymbol{\theta}) &= \sum_{k=1}^{K} E_{\boldsymbol{Z}_k}\left[\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)^T\right] \\ &= \sum_{k=1}^{K}\left[\sum_{Z_{k10}=0}^{m_T}\cdots\sum_{Z_{kT0}=0}^{m_T}\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)\left(\frac{\partial \ell_k(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right) Pr(\boldsymbol{Z}_k)^T\right] \end{aligned}$$

- Calculate $Var(\hat{\boldsymbol{\theta}}) = \boldsymbol{I}(\boldsymbol{\theta})^{-1}$, and $Var(\hat{\beta}^*)$ is the (2,2)-element of $Var(\hat{\boldsymbol{\theta}})$
- Power calculation

$$power = \Phi\left(\frac{|\beta_A|}{\sqrt{Var(\hat{\beta}^*)}} - Z_{1-\alpha/2}\right)$$

Gamma Distributions and Normal Distribution

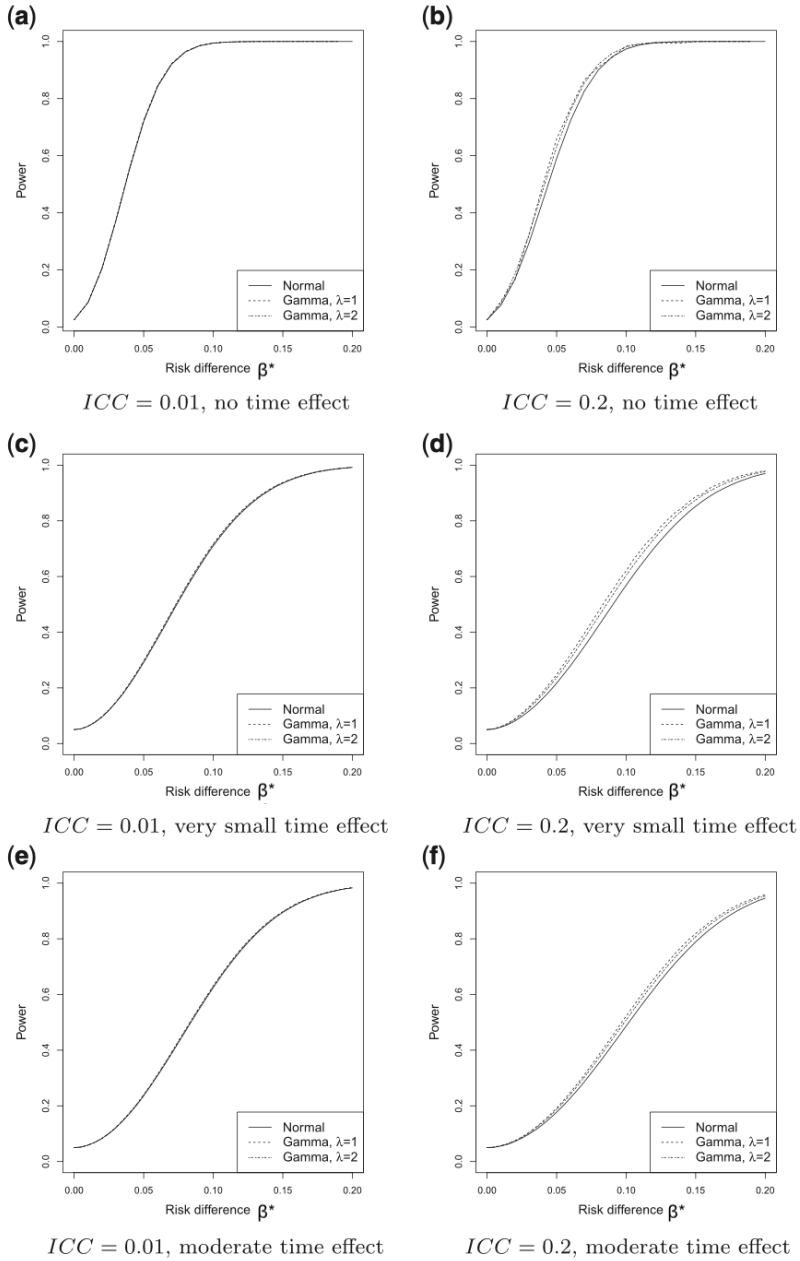
We now investigate the robustness of methods to departures from normality by considering two versions of the Gamma distribution



(a) Shape $\lambda = 1$ (b) Shape $\lambda = 2$

Fig. S2. Gamma and standard normal densities with the same mean and variance.

The robustness of methods to departures from normality by considering two versions of the Gamma distribution



$ICC = 0.01$, no time effect

$ICC = 0.2$, no time effect

$ICC = 0.01$, very small time effect

$ICC = 0.2$, very small time effect

$ICC = 0.01$, moderate time effect

$ICC = 0.2$, moderate time effect

Fig. 2. Power vs. the risk difference β^* for different cluster random effect distributions, with baseline risk $p_0 = 0.05$, number of steps $J = 3$, number of clusters $I = 8$, and cluster size $N = 90$. For figures in the left column, $ICC = 0.01$; while for figures in the right column, $ICC = 0.2$. There are no time effects ($\gamma_j = 0$) in the first row, very small time

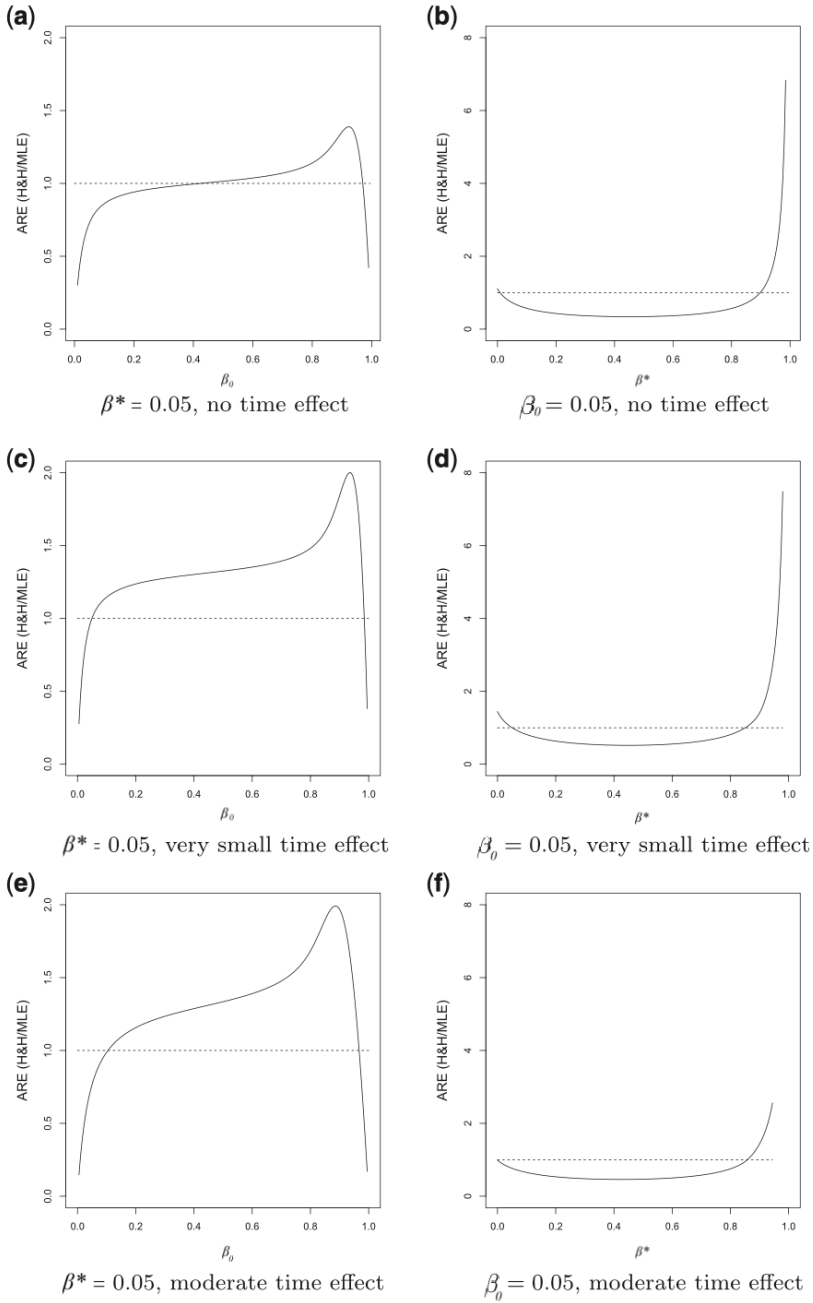
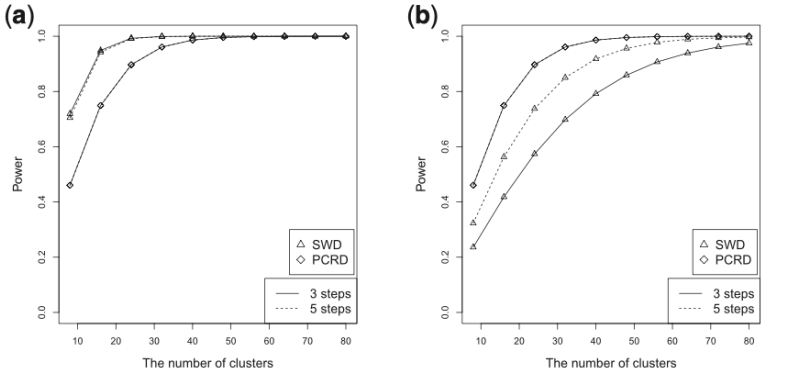
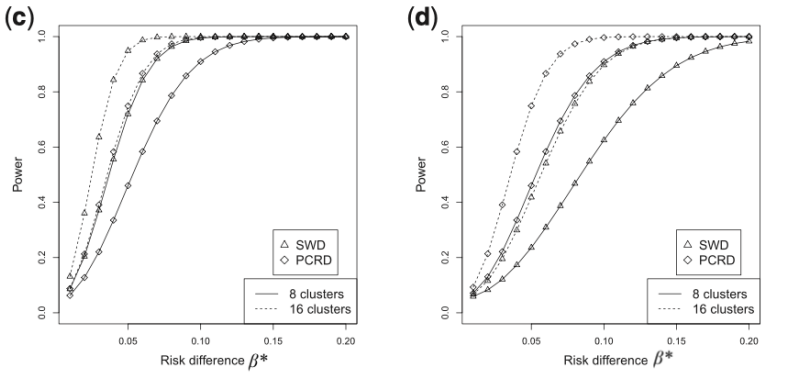


Fig. 3. ARE of $\widehat{\beta_{HH}}$ relative to $\widehat{\beta_{ML}}$. Figures in the left column show ARE vs. baseline risk β_0 with the risk difference $\beta_0 = 0.05$; Figures in the right column show ARE vs. risk difference β^* with the baseline risk $\beta_0 = 0.05$. There are no time effects ($\gamma_T = 0$) in the first row, very small time effects in the second row ($\gamma_T = 0.0001$), and moderate time effects in the third row ($\gamma_T = 0.05$). The number of clusters $I = 8$, the number of steps $J = 5$, the cluster size $N = 90$, and $ICC = 0.01$.



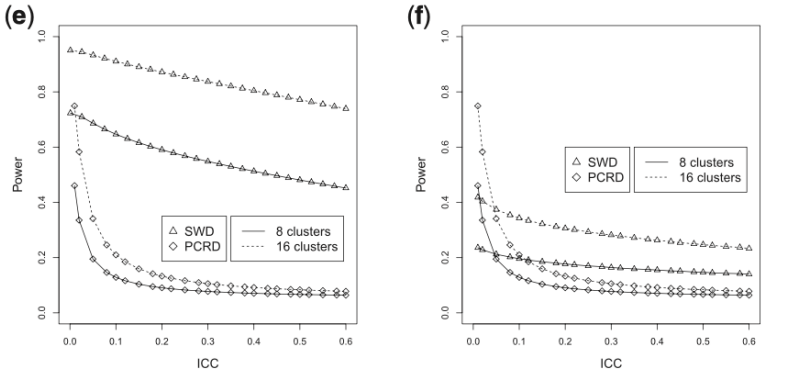
power vs. number of clusters, no time effect

power vs. number of clusters, moderate time effect



power vs. risk difference, no time effect

power vs. risk difference, moderate time effect



power vs. ICC with $\beta_0 = 0.05$, no time effect

power vs. ICC with $\beta_0 = 0.05$, moderate time effect

Fig. 4. Comparison between SWD and pCRD. There are no time effects in the model for figures in the left column, and moderate time effects in the model for figures in the right column. (a) and (b) power vs. the number of clusters, with baseline risk $\beta_0 = 0.05$, the risk difference $\beta^* = 0.05$, the cluster size $N = 90$, and $ICC = 0.01$; (c) and (d) power vs. the risk difference β^* with baseline risk $\beta_0 = 0.05$, the cluster size $N = 90$, the $ICC = 0.01$, and the number of steps $J = 3$; (e) and (f) power vs. ICC, for different baseline risks μ, where the risk difference $\beta^* = 0.05$, the cluster

Marginal Model – Generalized Estimating Equation Approach

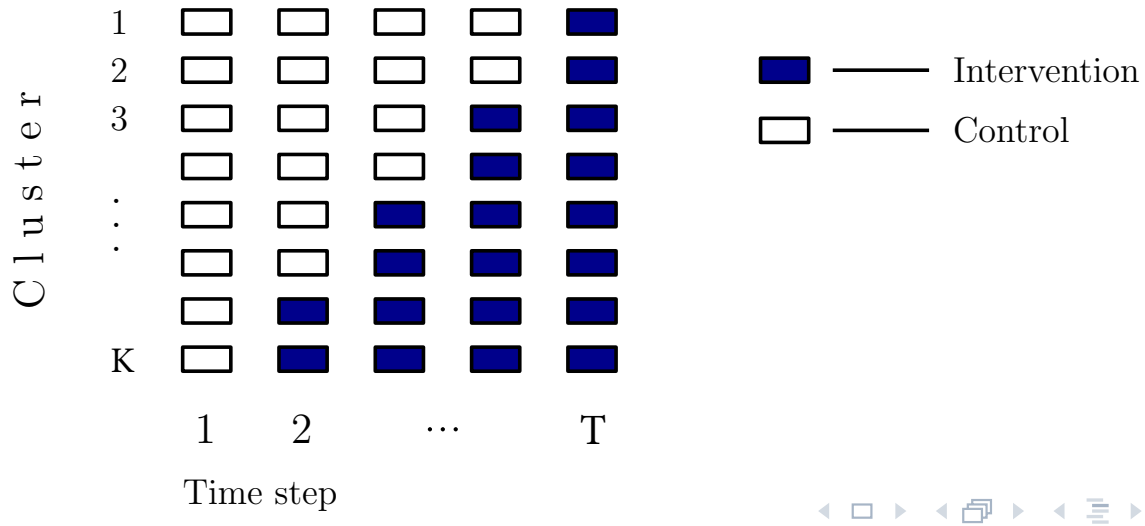
GEE model with time effects

- Apply a generalized estimating equations model (GEE) using the logit link for binary outcomes

$$p_{ktj} = g^{-1}(\beta_0 + X_{ktj}\beta^* + \gamma_t),$$

where γ_t is the time effect corresponding to step t ($\gamma_1 = 0$ for identifiability)

- $p_{ktj} = P(Y_{ktj} = 1 | X_{kt}, \gamma_t)$.
- The link function $g(\cdot)$ is typically identity, log and logit links.
- The parameters $\boldsymbol{\theta} = (\beta_0, \beta^*, \gamma_2, \cdots, \gamma_T)$.



Method – Generalized Estimating Equations

GEE model with time effects (Li et al, 2021[1]):

- For cluster k, $Y_k = (Y_{k11}, Y_{k12}, \cdots, Y_{kTm_T})^T$ and $p_k = (p_{k11}, p_{k12}, \cdots, p_{kTm_T})^T$.
- The GEE estimator $\hat{\theta}$ is obtained by solving the equation

$$\sum_{k=1}^{K} \mathrm{D}_k^T \mathrm{V}_k{}^{-1} (Y_k - p_k) = 0.$$

the empirical working variance-covariance

  - $\mathrm{D}_k = \partial p_k / \partial \theta$, $\mathrm{V}_k = A_k{}^{1/2} R_k A_k{}^{1/2}$
  - A_k is the $m-$dimensional diagonal matrix with elements $v(p_{ktj})$
  - The variance function for binary data is $v(p_{ktj}) = p_{ktj}(1 - p_{ktj})$.
  - R_k is the working correlation matrix for cluster k
- For study design purposes, the covariance of $\hat{\theta}$ is estimated by the model-based estimator $(\sum_k \mathrm{D}_k(\theta)^T \mathrm{V}_k^{-1}(\alpha) \mathrm{D}_k(\theta))^{-1}$. (The derivation is shown on a "GEE_variance.pdf" in the lecture 3 folder.)

  Hence, we can calculate the variance of $\hat{\beta}$.
- Comparison with Hussey & Hughes method (with the identity link)
  - Hussey & Hughes method assumes the variance of outcomes is fixed, and it is independent of the treatment and time step.
  - The GEE method uses the variance of Bernoulli distribution, which is a function of the treatment and time step effects.

[1] Li F et al (2021). Mixed-effects models for the design and analysis of stepped wedge cluster randomized trials: An overview. Statistical Methods in Medical Research, 30(2), 612-639.

Method – Generalized Estimating Equations

- The working correlation matrix R_k for cluster k for cross-sectional designs:

$$\mathrm{R}_k = (1-\alpha_0)\mathrm{I}_m + (\alpha_0 - \alpha_1)\mathrm{I}_T \otimes \mathrm{J}_{m_T} + \alpha_1 J_m,$$

  where J_u is a $u * u$ matrix with all elements of 1, I_u is a $u * u$ identity matrix.
- The working correlation matrix R_k for cluster k for cohort designs:

$$\mathrm{R}_k = (1-\alpha_0+\alpha_1-\alpha_2)\mathrm{I}_m + (\alpha_2 - \alpha_1)\mathrm{J}_T \otimes \mathrm{I}_{m_T} + (\alpha_0 - \alpha_1)\mathrm{I}_T \otimes \mathrm{J}_{m_T} + \alpha_1 J_m.$$

- A correlation structure with three levels of correlation is employed:
  - α_0, within-step correlation ($\mathrm{corr}(Y_{ktj}, Y_{ktj'}) = \alpha_0$, $j \neq j'$)
  - α_1, between-step correlation ($\mathrm{corr}(Y_{ktj}, Y_{kt'j'}) = \alpha_1$, $t \neq t', j \neq j'$)
  - α_2, within-individual correlation ($\mathrm{corr}(Y_{ktj}, Y_{kt'j}) = \alpha_2$, $t \neq t'$)

working empirical correlation

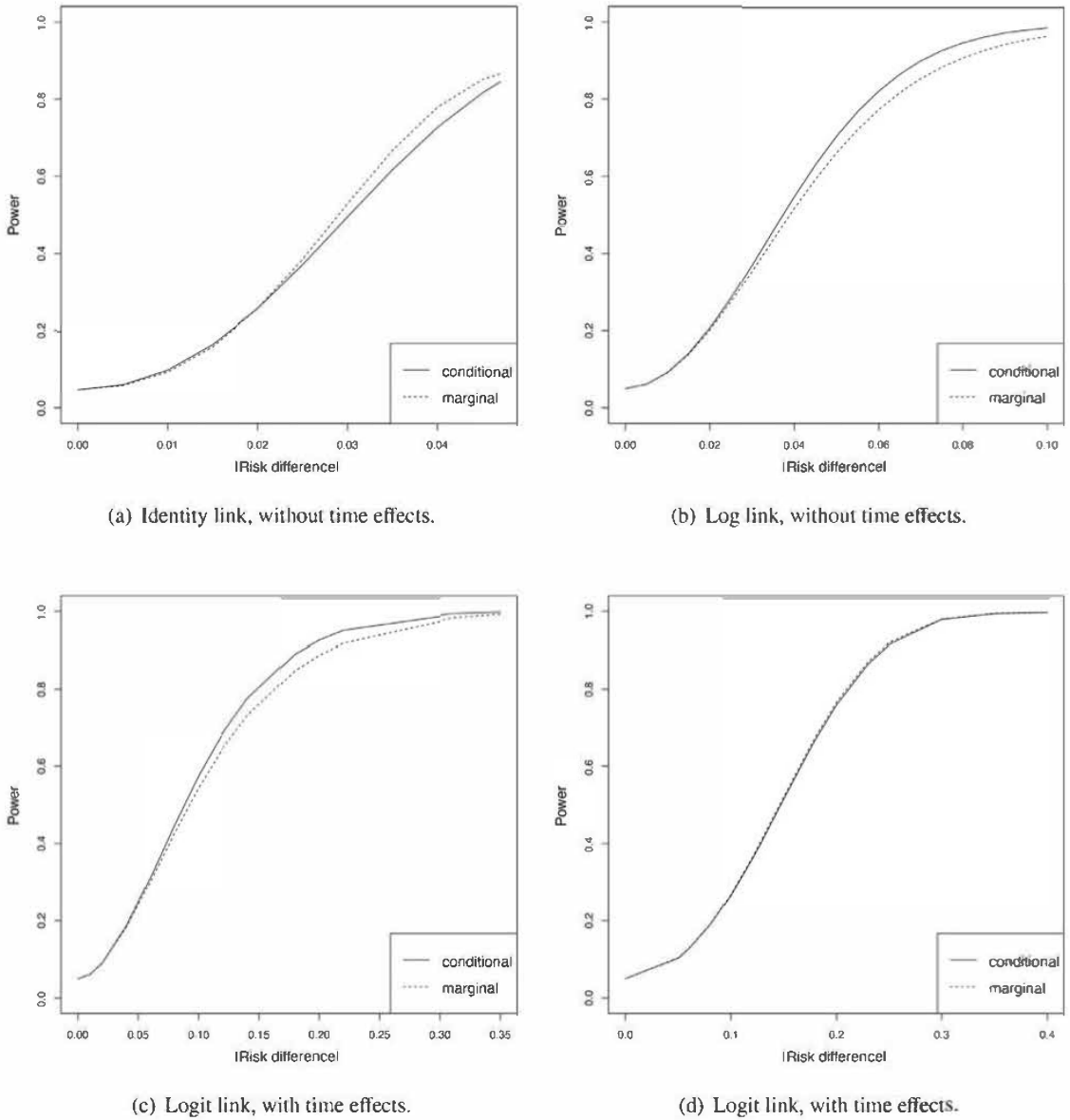
	Y_{111}	Y_{112}	Y_{113}	Y_{114}	Y_{121}	Y_{122}	Y_{123}	Y_{124}	Y_{131}	Y_{132}	Y_{133}	Y_{134}
Y_{111}	1	α_0	α_0	α_0	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_1
Y_{112}	α_0	1	α_0	α_0	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_1
Y_{113}	α_0	α_0	1	α_0	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_1
Y_{114}	α_0	α_0	α_0	1	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_1
Y_{121}	α_1	α_1	α_1	α_1	1	α_0	α_0	α_0	α_1	α_1	α_1	α_1
Y_{122}	α_1	α_1	α_1	α_1	α_0	1	α_0	α_0	α_1	α_1	α_1	α_1
Y_{123}	α_1	α_1	α_1	α_1	α_0	α_0	1	α_0	α_1	α_1	α_1	α_1
Y_{124}	α_1	α_1	α_1	α_1	α_0	α_0	α_0	1	α_1	α_1	α_1	α_1
Y_{131}	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_1	1	α_0	α_0	α_0
Y_{132}	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_0	1	α_0	α_0
Y_{133}	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_0	α_0	1	α_0
Y_{134}	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_1	α_0	α_0	α_0	1

(a) cross-sectional

	Y_{111}	Y_{112}	Y_{113}	Y_{114}	Y_{121}	Y_{122}	Y_{123}	Y_{124}	Y_{131}	Y_{132}	Y_{133}	Y_{134}
Y_{111}	1	α_0	α_0	α_0	α_2	α_1	α_1	α_1	α_2	α_1	α_1	α_1
Y_{112}	α_0	1	α_0	α_0	α_1	α_2	α_1	α_1	α_1	α_2	α_1	α_1
Y_{113}	α_0	α_0	1	α_0	α_1	α_1	α_2	α_1	α_1	α_1	α_2	α_1
Y_{114}	α_0	α_0	α_0	1	α_1	α_1	α_1	α_2	α_1	α_1	α_1	α_2
Y_{121}	α_2	α_1	α_1	α_1	1	α_0	α_0	α_0	α_2	α_1	α_1	α_1
Y_{122}	α_1	α_2	α_1	α_1	α_0	1	α_0	α_0	α_1	α_2	α_1	α_1
Y_{123}	α_1	α_1	α_2	α_1	α_0	α_0	1	α_0	α_1	α_1	α_2	α_1
Y_{124}	α_1	α_1	α_1	α_2	α_0	α_0	α_0	1	α_1	α_1	α_1	α_2
Y_{131}	α_2	α_1	α_1	α_1	α_2	α_1	α_1	α_1	1	α_0	α_0	α_0
Y_{132}	α_1	α_2	α_1	α_1	α_1	α_2	α_1	α_1	α_0	1	α_0	α_0
Y_{133}	α_1	α_1	α_2	α_1	α_1	α_1	α_2	α_1	α_0	α_0	1	α_0
Y_{134}	α_1	α_1	α_1	α_2	α_1	α_1	α_1	α_2	α_0	α_0	α_0	1

(b) closed cohort

Power vs. risk difference, by link function, with and time effects



(a) Identity link, without time effects.

(b) Log link, without time effects.

(c) Logit link, with time effects.

(d) Logit link, with time effects.

Appendix A.1. Power vs. the absolute value of risk difference between the control and intervention group at the end of the cross-sectional study, for K=6, m_T=4o, T=3, and

$\alpha_0 = \alpha_1 = 0.02$. There are no time effects in the first row. Fig. A.1a has the identity link and the baseline mean response in the control group of 0.05. The conditional model has a slightly larger power when the absolute value of risk difference is smaller, the opposite conclusion is shown when the absolute value of risk difference is larger. Fig. A.1b has the log link and the baseline mean response in the control group of 0.045. The conditional and marginal models have almost the same power when risk difference is smaller, and conditional model has a larger power when risk difference increases. The second row considers time effects and the logit link. In Fig. A.1c, the baseline mean response in the control group is 0.03 and the mean response in the control group at the end of the study is 0.04. The conditional model has a larger power as risk difference increases. In Fig. A.1d, the baseline mean response in the control group is 0.3 and the mean response in the control group at the end of the study is 0.4. The marginal model has a larger power as risk difference increases.

Example

PPIUD study

- Outcome: pregnancy within 1.5 years of the index birth
- 6 hospitals and 4 time steps
- m_T=900, cluster size $m = Tm_T$=3600
- Baseline pregnancy rate: 18.1%=β_0
- Expected pregnancy rate after intervention
  - 14.5% – relative risk 0.8
  - 16.3% – relative risk 0.9
- ICC ρ=0.022
- Hypothetical time effects
  - No time effects $\gamma_t = 0, t = 2, .., T$
  - Negligible time effects $\gamma_T = -1\% \times \beta_0 = -0.01 \times 0.181 = -0.00181$ and $\gamma_t = \gamma_T/(T-1), t = 2, ..., T$
  - 5% reduction of the baseline pregnancy rate $\gamma_T = -0.0091$
  - 10% reduction of the baseline pregnancy rate $\gamma_T = -0.0181$

Design example

Expected pregnancy rate	Time effects							
	$\gamma_T = 0$		γ_T=-0.181%		$\gamma_T = -0.91\%$		$\gamma_T = -1.81\%$	
	RR=0.8	RR=0.9	RR=0.8	RR=0.9	RR=0.8	RR=0.9	RR=0.8	RR=0.9
MLE (conditional)	1	0.908	0.976	0.480	0.981	0.494	0.984	0.506
GEE (marginal)	1	0.859	0.945	0.423	0957	0.449	0.967	0.474
Hussey & Hughes	1	0.850	0.935	0.412	0.935	0.412	0.935	0.412

Background parameters: $K = 6$, $T = 4$, $m_T = 900$, ICC$= 0.022$, baseline pregnancy rate $18.1\% = \beta_0$, RR$\in \{0.8, 0.9\}$, time effects $\delta \in \{0, -0.01\%, -0.91\%, -1.81\%\}$.

Software

- Hussey and Hughes method is an ad hoc approximation for binary outcomes.
- To improve this approximation, Zhou et al. proposed a conditional method, and Li et al. proposed a marginal method for binary outcomes, respectively, as discussed above.
- We have developed user-friendly software `swdpwr` to implement power calculations for SWDs with both continuous and binary outcomes and incorporate many possible scenarios.
- Chen J, Zhou X, Li F, Spiegelman D. swdpwr: A SAS macro and an R package for power calculations in stepped wedge cluster randomized trials. Comput Methods Programs Biomed. 2022 Jan;213:106522. doi: 10.1016/j.cmpb.2021.106522. Epub 2021 Nov 12.

Software

Two Statistical Models

- Only complete designs are supported. In complete designs, all clusters start from the first period, and end at the last period. Incomplete designs occur, for example, when there are transition periods between the control and the intervention. Another incomplete design would be where some clusters stop early in the study, and some clusters start late in the study.
  - Recall that in complete SWDs, the number of clusters (K) is equal to number steps ($T-1$) multiplied by the number of clusters switching at each step (R), that is, $K = (T-1) \times R$
- A correlation structure with three levels of correlation is employed:
  - α_0, within-period correlation of outcomes within clusters ($\text{corr}(Y_{ktj}, Y_{ktj'}) = \alpha_0$, $j \neq j'$)
  - α_1, between-period correlation of outcomes within clusters ($\text{corr}(Y_{ktj}, Y_{kt'j'}) = \alpha_1$, $t \neq t', j \neq j'$)
  - α_2, within-individual correlation across time periods ($\text{corr}(Y_{ktj}, Y_{kt'j}) = \alpha_2$, $t \neq t'$)
- As discussed in Lecture 3, for continuous outcomes, the marginal method and conditional method are equivalent. In both models, the covariance is estimated by the model-based estimator $(\sum_k X_k^T V_k^{-1} X_k)^{-1}$, where X_k is the $m * (T+1)$ design matrix corresponds to the parameter vector and $V_k = \sigma_t^2 R_k$ within cluster k.
- Our software supports **cross-sectional and cohort** SWDs with continuous and binary outcomes.

Software

Conditional Model - Binary Outcomes

For binary outcomes, consider a **cross-sectional** SWD which is modelled by a generalized linear mixed model (GLMM):

$$g(p_{ktj}) = g(\beta_0 + X_{kt}\beta^* + \gamma_t + b_k). \tag{3}$$

- $g(\cdot)$ can choose from: identity, log and logit links.
- $b_k \sim N(0, \sigma_B^2)$ is the random effect.
- The correlation structure for this case is $\alpha_0 = \alpha_1 = \alpha_2$. **\alpha_2 = 0**
- $p_{ktj} = P(Y_{ktj} = 1 | X_{kt}, b_k)$ is the conditional probability of the outcome for each individual.
- Then we derive the likelihood of the observed outcomes based on the conditional probability under the specific link function. The calculation of the large-sample variance and power is then based on the theory of MLE.
- We also consider the special case of SWDs with no time effects (all $\gamma_t = 0$). The model is

$$g(p_{ktj}) = g(\beta_0 + X_{kt}\beta^* + b_k).$$

The derivations of the likelihood formula and power are similar to the procedures with time effects. Methods follow Zhou et al. (SIM, 2020)

Software

Marginal Model - Binary Outcomes

For binary outcomes, we assume a **cross-sectional or cohort** SWD modelled by a generalized estimating equations (GEE) model:

$$p_{ktj} = g^{-1}(\beta_0 + X_{ktj}\beta^* + \gamma_t),$$

- $g(\cdot)$ can choose from: identity, log and logit links.
- $p_{ktj} = P(Y_{ktj} = 1|X_{kt})$ is the marginal probability of the outcome.
- The parameters $\boldsymbol{\theta} = (\beta_0, \beta^*, \gamma_2, \cdots, \gamma_T)$.
- The GEE estimator $\hat{\boldsymbol{\theta}}$ is obtained by solving the equation

$$\sum_k \mathrm{D}_k^T \mathrm{V}_k{}^{-1}(Y_k - p_k) = 0.$$

  - $\mathrm{D}_k = \partial p_k/\partial\boldsymbol{\theta}$, $\mathrm{V}_k = A_k{}^{1/2} R_k A_k{}^{1/2}$
  - A_k is the $m-$dimensional diagonal matrix with elements $v(p_{ktj})$
  - The variance function is $v(p_{ktj}) = p_{ktj}(1 - p_{ktj})$.
  - The working correlation matrix for cluster k :

$$\mathrm{R}_k = (1 - \alpha_0 + \alpha_1 - \alpha_2)\mathrm{I}_m + (\alpha_2 - \alpha_1)\mathrm{J}_T \otimes \mathrm{I}_{m_T} + (\alpha_0 - \alpha_1)\mathrm{I}_T \otimes \mathrm{J}_{m_T} + \alpha_1 J_m.$$

- The covariance of $\hat{\boldsymbol{\theta}}$ is estimated by the model-based estimator $(\sum_k \mathrm{D}_k(\boldsymbol{\theta})^T \mathrm{V}_k^{-1}(\alpha)\mathrm{D}_k(\boldsymbol{\theta}))^{-1}$. From this expression, the variance of $\hat{\beta}^*$ is derived.

Software

Methods implemented (Chen et al, 2022):

		Conditional model	**Marginal model**
Binary outcomes (identity, log, logit links)	Cross-sectional design	Zhou et al., 2020 (exchangeable structure)	Li et al., 2018a (nested exchangeable structure)
	Cohort design	N/A	Li et al., 2018a (block exchangeable structure)
Continuous outcomes (identity link)	Cross-sectional design	Hussey and Hughes, 2007; Li et al., 2018a; Zhou et al., 2017 (nested exchangeable structure)	
	Cohort design	Hooper et al.,2016; Li et al., 2018b; Li et al., 2018a (block exchangeable structure)	

The arguments of swdpower in the R package swdpwr

```
swdpower(K, design, family = "binomial", model = "conditional", link = "identity",
type = "cross-sectional", meanresponse_start = NA, meanresponse_end0 = meanresponse_start,
meanresponse_end1 =  NA, effectsize_beta = NA, sigma2 = 0, typeIerror = 0.05, alpha0 = 0.1,
alpha1 = alpha0/2, alpha2 = NA)
```

Note: K in this software is m_T in the lecture notes.

Software

Argument	Description	Default
`K`	Number of individuals at each time period in a cluster, or the average cluster-period size under variable cluster-period sizes.	User input required, no default (an error is produced if this parameter isn't supplied.)
`design`	`I*J` dimensional data set that describes the study design, unequal allocation of sequences, and only complete designs with no transition periods allowed.	User input required, no default (an error is produced if this parameter isn't supplied.)
`family`	Specify `family="gaussian"` for continuous outcome, `family="binomial"` for binary outcome.	`"binomial"`
`model`	Specify `model="conditional"` for conditional model, `model="marginal"` for marginal model.	`"conditional"`
`link`	Choose link function from `link="identity"`, `link="log"`, and `link="logit"`.	`"identity"`
`type`	Specify `type="cohort"` for closed cohort study, `type="cross-sectional"` for cross-sectional study.	`"cross-sectional"`
`meanresponse_start`	The anticipated mean response in the control group at the start of the study.	User input required, no default (an error is produced if this parameter isn't supplied.)
`meanresponse_end0`	The anticipated mean response in the control group at the end of the study.	`meanresponse_start` (effectsize_beta = meanresponse_end1 - meanresponse_end0)
`meanresponse_end1`	The anticipated mean response in the intervention group at the end of the study.	User can choose to specify meanresponse_start and meanresponse_end1 without specifying effectsize_beta, or alternatively specify meanresponse_start and effectsize_beta.
`effectsize_beta`	The anticipated effect size.	User can choose to specify meanresponse_start and meanresponse_end1 without specifying effectsize_beta, or alternatively specify meanresponse_start and effectsize_beta.
`sigma2`	Marginal variance of the outcome (only needed if it is a continuous outcome).	`0`
`typeIerror`	Two-sided Type I error.	`0.05`
`alpha0`	Within-period intracluster correlation α_0.	`0.1`
`alpha1`	Between-period intracluster correlation α_1.	`alpha0/2`
`alpha2`	Within-individual correlation α_2 (only allowed if `type="cohort"`).	Required for option "cohort", otherwise, not used.

Software

Example 1 (conditional method for binary outcome and cross-sectional design)

An investigator plans to conduct a cross-sectional SWD with 12 clusters and 3 time steps.

- This trial has an average of 50 individuals per cluster per period with a total sample size of 1800, and measurements of the binary outcome will be taken.
- The baseline proportion is assumed to be 0.2 and is expected to increase to 0.25 by the end of the study even in the absence of the intervention, due to the secular trend.
- The risk difference by the end of the study is specified to be 0.13.
- The investigator assumes a constant ICC over time, equal to 0.01.
- The treatment effect is to be tested at a two-sided 5% level of significance.

Software

```
R> library("swdpwr")
R> data = matrix(c(rep(c(0, 1, 1), times=6),
+ rep(c(0, 0, 1), times=6)), nrow=12, ncol=3, byrow = TRUE)
R> data
      [,1] [,2] [,3]
 [1,]    0    1    1
 [2,]    0    1    1
 [3,]    0    1    1
 [4,]    0    1    1
 [5,]    0    1    1
 [6,]    0    1    1
 [7,]    0    0    1
 [8,]    0    0    1
 [9,]    0    0    1
[10,]    0    0    1
[11,]    0    0    1
[12,]    0    0    1
R> swdpower(K = 50, design = data, family = "binomial",  model = "conditional",
+ link = "identity", type = "cross-sectional", meanresponse_start = 0.2,
+ meanresponse_end0 = 0.25, meanresponse_end1 = 0.38, typeIerror = 0.05,
+ alpha0 = 0.01, alpha1 = 0.01)

This cross-sectional study has total sample size of 1800
Power for this scenario is 0.899 for the alternative hypothesis treatment effect
beta = 0.13 ( two-sided Type I error =  0.05 )
```

Software

Example 2 (marginal method for binary outcome and cohort design)

An investigator plans to conduct a closed cohort SWD with 12 clusters and 4 time steps. This trial has an average of 50 individuals per cluster with a total sample size of 600, and repeated measurements of the binary outcome will be taken at each time step for each individual. The baseline proportion is assumed to be 0.156 and is expected to increase to 0.1765 by the end of the study even in the absence of the intervention (due to the secular trend). The relative risk by the end of the study is specified to be $e^{0.5} = 1.65$. The investigator assumes a within-step ICC of 0.03, a between-step ICC of 0.015, and a within-individual correlation in repeated measures on the same individual of 0.2. The treatment effect is to be tested at a two-sided 5% level of significance.

Software

```
R> data = matrix(c(rep(c(0, 1, 1, 1), times=4),rep(c(0, 0, 1,1), times=4),
+rep(c(0, 0, 0, 1), times=4)),nrow=12, ncol=4, byrow = TRUE)
R> data
      [,1] [,2] [,3] [,4]
 [1,]    0    1    1    1
 [2,]    0    1    1    1
 [3,]    0    1    1    1
 [4,]    0    1    1    1
 [5,]    0    0    1    1
 [6,]    0    0    1    1
 [7,]    0    0    1    1
 [8,]    0    0    1    1
 [9,]    0    0    0    1
[10,]    0    0    0    1
[11,]    0    0    0    1
[12,]    0    0    0    1
R> swdpower(K = 50, design = data, family = "binomial",  model = "marginal",
+ link = "log", type = "cohort", meanresponse_start = 0.156,
+ meanresponse_end0 = 0.1765, effectsize_beta = 0.5, typeIerror = 0.05,
+ alpha0 = 0.03, alpha1 = 0.015, alpha2 = 0.2)

This cohort study has total sample size of 600
Power for this scenario is 0.819 for the alternative hypothesis treatment effect
beta = 0.5 ( two-sided Type I error =  0.05 )
```

Software

Example 3 (continuous outcome and cohort design)

An investigator plans to conduct a closed cohort SWD with 8 clusters and 3 time steps. This trial has an average of 24 individuals per cluster with a total sample size of 192, and repeated measurements of the continuous outcome will be taken at each time step for each individual. The baseline mean response is assumed to be 0.1 and is expected to increase to 0.2 by the end of the study even in the absence of the intervention (due to the secular trend). The mean response difference by the end of the study is specified to be 0.2. The investigator assumes a within-step ICC of 0.03, a between-step ICC of 0.015, and a within-individual correlation in repeated measures on the same individual of 0.2. The marginal variance of the outcome is set to 0.095. The treatment effect is to be tested at a two-sided 5% level of significance.

Software

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R> data = matrix(c(rep(c(0, 1, 1), times = 4),
+ rep(c(0, 0, 1), times = 4)), nrow = 8, ncol = 3, byrow = TRUE)
R> data
     [,1] [,2] [,3]
[1,]    0    1    1
[2,]    0    1    1
[3,]    0    1    1
[4,]    0    1    1
[5,]    0    0    1
[6,]    0    0    1
[7,]    0    0    1
[8,]    0    0    1
R> swdpower(K = 24, design = data, family = "gaussian",  model = "marginal",
+ link = "identity", type = "cohort", meanresponse_start = 0.1,
+ meanresponse_end0 = 0.2, effectsize_beta = 0.2, sigma2 = 0.095,
+ typeIerror = 0.05, alpha0 = 0.03, alpha1 = 0.015, alpha2 = 0.2)

This cohort study has total sample size of 192
Power for this scenario is 0.965 for the alternative hypothesis treatment effect
beta = 0.2 ( two-sided Type I error =  0.05 )
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References

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List of Notation

Y_{ktj}: outcome of individual $j(\in \{1, \cdots, m_T\})$ at step $t(\in \{1, \cdots, T\})$ from cluster $k(\in \{1, \cdots, K\})$
X_{ktj}: an indicator of the treatment (1=intervention; 0=control)
β_0: the expectation of the outcome at baseline (control at step 1)
β^*: the treatment effect
γ_t: the time effect corresponding to step t ($\gamma_1 = 0$ for identifiability)
$\epsilon_{ktj} \sim N(0, \sigma_W^2)$: individual-level error term
$b_k \sim N(0, \sigma_B^2)$: random effect for cluster k
p_{ktj}: $= P(Y_{ktj} = 1|X_{kt})$
α_0: within-period correlation between clusters ($\text{corr}(Y_{ktj}, Y_{ktj'}) = \alpha_0$, $j \neq j'$)
α_1: inter-period correlation between clusters ($\text{corr}(Y_{ktj}, Y_{kt'j'}) = \alpha_1$, $t \neq t', j \neq j'$)
α_2: within-individual correlation ($\text{corr}(Y_{ktj}, Y_{kt'j}) = \alpha_2$, $t \neq t'$)
$c_{kt} \sim N(0, \sigma_c^2)$: the cluster-by-time interaction random effect, typically assumed to be 0 at the design stage
$\pi_{kj} \sim N(0, \sigma_\pi^2)$: the random effect for repeated measures of one individual.
$\boldsymbol{U}(\boldsymbol{\theta}) = \frac{\partial \ell(\boldsymbol{\theta};\boldsymbol{y})}{\partial \boldsymbol{\theta}}$: score function
$\boldsymbol{I}(\boldsymbol{\theta}) = E(\boldsymbol{U}(\boldsymbol{\theta})\boldsymbol{U}(\boldsymbol{\theta})^T)$: information matrix